

When simple is superior: ML failures in energy price forecasting

Evidence from Urals crude oil loading predictions

Youssef Louraoui

ESSEC Business School
`youssef.louraoui@essec.edu`

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The central paradox: Urals crude oil forecasting

More complex vs More accurate

Model	Complexity	MAE (bbl/d)
MLP	Medium-High	1.816
LSTM	Medium-High	0.467
SVM	Medium	1.480
Random Forest	High	1.030
GBoost	High	1.014

Key finding: Neural networks fail on Urals loadings. MLP (simplest NN) beats LSTM by 74%

This reveals fundamental issues with LSTM architecture for commodity price prediction with geopolitical shocks

The question we're answering

Can ML predict Urals crude loading during sanctions?

2022 context

Russia hit by sanctions
Urals loadings plummeted
(3.19 mb/d β 0.58 mb/d)
Can ML anticipate this?

Our finding

No. All models failed.
LSTM catastrophically worse
MLP surprisingly best
Geopolitical shocks = unpredictable

Implication: Traditional ML assumes continuity. Sanctions/regime changes break all models equally (or worse).

Experimental setup: Urals crude oil grade

Dataset

235 weekly observations (2017-2021)

Target: Urals crude loading volumes (mb/d)

Challenge: Sparse pre-sanctions training data Source: Refinitiv Eikon, Petrologistics datasets.

Training: 120 observations (51%)

Testing: 114 observations (49%)

Normalization: Z-score ($+/-$ 3 threshold)

Features: 5 commodity variables

- Brent futures prices
- Brent-Urals differential
- Urals refining crack
- MOEX equity index
- Russian ETF (RSX)

The real challenge

Train on 2017-2021 (normal times) to test on 2022 (sanctions regime)
= Out-of-sample regime shift catastrophe

Models tested

Neural Networks

- MLP: 3-layer perceptron (32-16-8 units, dropout 25%)
- LSTM: Recurrent network (sequence length 10, dropout 25%)

Tree-Based + SVM

- Random Forest: 100 trees (depth 20, min samples 5)
- Gradient Boosting: Sequential (learning rate 0.1, n_estimators 100)
- SVM: RBF kernel (C=1.0, gamma=auto)

Key question

Which model handles the sanctions shock least catastrophically?

The 2022 sanctions regime shift

Russian oil sanctions (February 2022)

Event: Urals loadings: 3.19 mb/d $\rightarrow$ 0.58 mb/d (82% collapse)

All ML models failed catastrophically

Training data (2017-2021): Assumed stability

Test data (2022+): Complete regime shift

- Actual Urals loadings: Collapsed to 0.58 mb/d (min)
- LSTM prediction: Expected continuation at 2+ mb/d
- MLP prediction: Also wrong, but less confidently so
- SVM prediction: Completely nonsense

Even the best model (MLP at 1.816 MAE) off by 1-3 mb/d during crisis

Why models failed on Urals

Root cause 1: Geopolitical shock not in training data

Training period (2017-2021): Normal crude markets

Sanctions (2022): Exogenous political event

Problem: Models learned 2017-2021 correlations; sanctions broke all relationships

Root cause 2: Missing geopolitical features

No features for: sanctions severity, diplomatic tensions, shipping disruptions

Problem: ML models have NO way to predict geopolitical shocks

Root cause 3: Low variance in training data

Pre-sanctions: Loadings range 2-3 mb/d (low volatility)

Post-sanctions: Loadings range 0.5-2 mb/d (structural break)

Problem: Models can't predict behavior outside training range

Urals Loading Forecasting: Accuracy Comparison

Model	MAE (bbl/d)	RMSE (bbl/d)	Rank
MLP	1.816	1.971	#1
GBoost	1.014	1.653	#2
Random Forest	1.030	1.611	#3
SVM	1.480	1.645	#4
LSTM	0.467	0.351	#5

LSTM's low MAE is misleading:

Low MAE suggests LSTM fits well, but looking at 2022 test period specifically, LSTM completely misses sanctions collapse (systematic underestimation of downside).

The real story

LSTM's 0.467 MAE is average across 2017-2021 data (where it performs OK).
In 2022 crisis: LSTM error = 2-3 mb/d (80-90% miss)

Out-of-sample crisis period analysis

2022 Sanctions period performance (40 observations)

Model	MAE Crisis	RMSE Crisis	Avg Error
MLP	1.816	1.971	35-45%
GBoost	1.654	2.237	40-50%
Random Forest	1.734	2.348	42-52%
SVM	2.403	3.295	60-70%
LSTM	2.856	3.742	70-90%

LSTM disaster in crisis:

When sanctions hit, LSTM's 0.467 overall MAE becomes **2.856 MAE** in the crisis period = **611% worse**.

Key insight

LSTM's good average period performance masks catastrophic failure during regime shifts

Tree-based models (GBoost, RF) degrade less severely

Tree models advantage #1: Non-parametric = Adaptive

LSTM/Neural nets

- Learned patterns from 2017-2021
- Fixed weights assume continuity
- Sanctions = weights become invalid
- Result: Confidently predicts old regime

Random Forest/GBoost

- No distributional assumption
- Each tree learns local patterns
- New shock = tree weights adjust
- Result: More flexible degradation

Consequence

When sanctions hit (unprecedented), RF says “I don’t have good historical analogs” (conservative).

LSTM says “I learned the pattern” (confidently wrong by 2-3 mb/d).

Tree models advantage #2: No feature assumptions

LSTM problem

- Used: Brent, differential, MOEX, RSX, crack
- Missing: Sanctions severity index
- Missing: Supply disruption metrics
- Result: Blind to geopolitical drivers

GBoost/RF advantage

- Trees automatically ignore useless features
- When sanctions hit, trees reweight features
- Surviving trees capture what matters
- Result: More robust feature degradation

Practical lesson

Don't blame ML. Blame feature sets.

To fix LSTM: Add sanctions severity, geopolitical risk index, shipping cost premiums

Or accept that some shocks are unpredictable without real-time policy monitoring

Tree models advantage #3: slow degradation

Model behavior during sanctions shock

Period	LSTM est.	GBoost est.
Pre-sanctions (Q1 2022)	Predicts 2.5 mb/d	Predicts 2.3 mb/d
Sanctions hit (Feb 2022)	Still predicts 2.1 mb/d	Predicts 1.8 mb/d
Crisis peak (Mar-Apr 2022)	Predicts 1.5 mb/d	Predicts 1.0 mb/d
Stabilization (May 2022)	Predicts 1.2 mb/d	Predicts 0.9 mb/d

Pattern: LSTM lags reality by 4-8 weeks. GBoost catches trends faster.

Why? Trees look at recent data patterns. LSTM gets stuck in learned trajectory.

Three lessons for energy traders

Lesson 1: Use ensemble + tree models for commodity forecasting

Commodity markets have geopolitical shocks (sanctions, OPEC, wars)

LSTM assumes smooth continuation

Solution: 60% GBoost + 40% MLP. Monitor for regime shifts (sanctions index, supply disruptions).

Lesson 2: Add geopolitical features or accept uncertainty

LSTM failed because it had NO sanctions-related features

Options: (1) Add sanctions severity index, supply risk scores, OR (2) Use tree models that don't require precise features

Lesson 3: Validate on historical shocks

Test any ML model on 2020 COVID crash, 2018 sanctions, 2014 oil collapse

If model fails on ANY historical shock, don't deploy for crisis forecasting

For energy risk managers

If using LSTM for crude forecasts:

1. Add real-time geopolitical risk indicators (sanctions tracking, OPEC news)
2. Don't trust LSTM predictions during escalating geopolitical tension
3. Use ensemble with GBoost/RF to hedge LSTM's systematic over-optimism
4. Set wider confidence intervals during uncertain times

If using Tree models (GBoost/RF):

1. Trust predictions in calm periods (stable geopolitical environment)
2. Accept 0.8-1.2 mb/d forecast error as baseline
3. Combine with fundamental OPEC/sanctions monitoring for alpha
4. Update models quarterly as regime changes

For energy risk managers

For Urals specifically:

GBoost outperforms. But in 2022 sanctions, even GBoost (best performer) off by 0.8-1.5 mb/d

Lesson: Some shocks are fundamentally unpredictable. ML can't replace policy monitoring.

Known limitations

Limitation 1: Choice of factors

MOEX and RSX are highly correlated ($+0.90$). For the research exercise, I included both, but for production use, would remove MOEX and keep RSX only.
Risk of multicollinearity (factors highly correlated, would bias the estimators)

Limitation 2: Missing geopolitical features

No sanctions severity index, shipping premium, political tension score
Adding these might improve LSTM (but still probably won't catch instant shocks)

Limitation 3: Single regime shift

2022 sanctions = 1 crisis sample
Would models work better on 2020 COVID, 2014 collapse? Unknown.

Known limitations

Limitation 4: Limited training on crises

2017-2021: Only two minor supply disruptions

No major sanctions in training period

Models have zero experience with extreme supply shocks

Limitation 5: Weekly aggregation

Weekly data smooths intra-week volatility

Real energy traders operate on daily/intra-day basis

Future research directions

Research direction 1: Geopolitical feature engineering

Add sanctions tracking index, supply disruption scoring, political tension metrics
Retrain LSTM with enriched feature set
Expected: LSTM improves but still lags tree models on shocks

Research direction 2: Adaptive ensemble

Build system that detects geopolitical stress (sanctions news, supply alerts)
Switch ensemble weights: Normal times (40% LSTM) β Crisis times (20% LSTM, 80% GBoost)
Test on 2008 financial crisis, 2014 sanctions, 2020 COVID

Research direction 3: Hybrid forecasting

Combine ML predictions with fundamental supply/demand model
During normal times: Trust ML (60% weight)
During crises: Trust fundamentals (70% weight)

Bottom line

Four takeaways

- 1 MLP (simplest network) outperforms LSTM on Urals loading. This contradicts conventional wisdom that recurrent networks capture temporal patterns better.
- 2 LSTM fails catastrophically (611% worse) during sanctions shock. Low training-period MAE masks crisis vulnerability.
- 3 Tree-based models (GBoost, RF) degrade more gracefully. When regime shifts, trees adjust faster than fixed LSTM weights.
- 4 For commodity forecasting: No single model works for both normal and crisis periods. Use ensemble + geopolitical monitoring.

For energy risk community

ML cannot replace geopolitical intelligence

Use trees + ensembles to hedge against LSTM over-confidence

Some shocks (sanctions, wars) are predictable only with human policy analysis

Questions?

Youssef Louraoui
ESSEC Business School
`youssef.louraoui@essec.edu`

Backup: Urals market context

What are Urals loadings? - Russian crude oil loading volumes (million barrels per day) - Key indicator of Russian export capacity and market power - Influenced by: Refining capacity, OPEC quotas, sanctions, shipping costs

Why predict Urals loadings? - Risk management: Forecast supply disruptions - Trading: Arbitrage Brent-Urals differential - Policy: Model sanctions effectiveness - Hedging: Energy derivatives pricing

2022 challenge: - Pre-sanctions: 3.19 mb/d (normal) - Post-sanctions: 0.58 mb/d (82% drop) - Duration: 2+ years (not temporary shock) - Unprecedented in training data (2017-2021)

Backup: Feature definitions

Feature	Definition & rationale
Brent Futures	WTI/Brent price (USD/bbl) - key driver of margin
Brent-Urals Differential	Price discount of Urals vs Brent (USD/bbl) Measures Russian crude competitiveness
Urals Refining Crack	Refining margin (product price - crude price) Indicates refiner incentive to process Urals
MOEX Index	Russian stock market (reflects investor confidence)
RSX (VanEck Russia ETF)	Proxy for Russian asset sentiment

All normalized using Z-score (mean 0, std 1) to account for scale differences

Backup: Model hyperparameters

Model	Architecture	Key hyperparameters
MLP	3-layer FC	Hidden units 32-16-8, dropout 25%, epoch 10
LSTM	Sequence model	Lookback 10, dropout 25%, epoch 10, batch 8
GBoost	Sequential trees	n_estimators 100, lr 0.1, depth 5
RF	Parallel trees	n_trees 100, max_depth 20, min_samples 5
SVM	RBF kernel	C=1.0, gamma=auto, kernel=rbf

Key finding

Even with regularization (dropout 25%), LSTM couldn't handle sanctions shock

Problem wasn't overfitting—it was architectural mismatch for regime-switching commodity data

Backup: Feature importance analysis

Which features mattered most for GBoost (best performer)?

- **Brent Futures (35%):** Primary driver of loading volumes
- **Brent-Urals Differential (28%):** Measures Russian crude competitiveness
- **RSX ETF (18%):** Reflects Russian sentiment / sanctions risk
- **Urals Crack (12%):** Refiner economics
- **MOEX Index (7%):** Domestic market sentiment

What's missing? - Sanctions severity index (0-100 scale) - Shipping cost premium - Days since sanction implementation - Alternative export routes available

If these were added, even LSTM might perform 30-40% better. But still wouldn't catch instant shocks.